

A Compartmental Model for Zika Virus with Dynamic Human and Vector Populations

Eva K Lee, PhD^{1,2,3,*}, Yifan Liu, BS^{1,2,3}, Ferdinand H Pietz, MPH⁴

¹NSF-Whitaker Center for Operations Research in Medicine and HealthCare; ²NSF I/UCRC Center for Health Organization Transformation; ³School of Industrial and Systems Engineering, Georgia Institute of Technology, Atlanta, GA. ⁴Strategic National Stockpile, Centers for Diseases Control and Prevention, Atlanta, GA.

Abstract

The Zika virus (ZIKV) outbreak in South American countries and its potential association with microcephaly in newborns and Guillain-Barré Syndrome led the World Health Organization to declare a Public Health Emergency of International Concern. To understand the ZIKV disease dynamics and evaluate the effectiveness of different containment strategies, we propose a compartmental model with a vector-host structure for ZIKV. The model utilizes logistic growth in human population and dynamic growth in vector population. Using this model, we derive the basic reproduction number to gain insight on containment strategies. We contrast the impact and influence of different parameters on the virus trend and outbreak spread. We also evaluate different containment strategies and their combination effects to achieve early containment by minimizing total infections. This result can help decision makers select and invest in the strategies most effective to combat the infection spread. The decision-support tool demonstrates the importance of “digital disease surveillance” in response to waves of epidemics including ZIKV, Dengue, Ebola and cholera.

*Corresponding author: eva.lee@gatech.edu

Introduction

Zika virus (ZIKV) is an arbovirus belonging to the family *Flaviviridae*, and is closely related to the dengue, yellow fever, Japanese encephalitis, and West Nile viruses. It was first isolated from a monkey in the Zika forest of Uganda in 1947¹. In 1948, the second isolation was made from *Aedes africanus* mosquitoes in the same forest^{1, 2}. Although earlier studies have suggested the possibility of human infection^{3, 4}, before 2007, ZIKV rarely caused recognized 'spillover' infections in humans, even in highly enzootic areas. Since 1951, human ZIKV infections have been reported in African countries including Uganda, Nigeria, Senegal, and Gabon, and parts of South Asia including India, Malaysia, and Indonesia^{5, 6, 7, 8}. In 2007, the first documented ZIKV outbreak was reported from Yap State, Federated States of Micronesia⁶. French Polynesia also reported an outbreak with 28,000 estimated cases in 2013⁹. Since April 2015, Brazil has been experiencing a ZIKV outbreak. The outbreak has subsequently spread to other countries in South America, Central America, and the Caribbean^{10, 11, 12}.

ZIKV is primarily transmitted by the bites of infectious *Aedes* mosquitoes¹³, especially *Aedes aegypti* and *Aedes albopictus*. Non-vector borne transmissions, though rare, have also been reported^{14, 15}. Symptoms of ZIKV infection include fever, rash, conjunctivitis, and retro-orbital pain. These symptoms are usually mild and unspecific¹⁶. The ZIKV infection is difficult to detect since its symptoms are similar to those of dengue and chikungunya¹³. Asymptomatic cases are also frequent, which makes it even harder to precisely diagnose the ZIKV infection^{13, 17, 18}. Although the symptoms of ZIKV infections are usually mild and unlethal, it has recently been connected to microcephaly in infants born to mothers infected with ZIKV during pregnancy, suggesting the possibility of intrauterine infection^{19, 20}. ZIKV is also suspected to be linked to Guillain-Barré Syndrome (GBS)²¹, a muscle weakness caused by the immune system damaging the peripheral nervous system.

With the ease of modern transportation, there is a potential that the ZIKV outbreak may become a global public health crisis.^{22, 23, 24}. Therefore, it is crucial to understand the transmission dynamics of ZIKV and establish a set of strategies that help to contain the outbreak effectively. Mathematical models have long been used to analyze infectious disease outbreaks to facilitate development of policies and strategies for understanding disease propagation and containment strategies^{25, 26, 27, 28}. Since ZIKV is closely related to West Nile, dengue, and yellow fever viruses²⁹, it can be modeled with similar techniques. Because of dengue's global impact, its disease dynamics have been studied extensively, and compartmental models based on Susceptible, Infectious, Recovered (SIR), and Susceptible, Exposed, Infectious, Recovered (SEIR) hosts have been proposed and analyzed^{30, 31, 32, 33, 34, 35}. The population dynamics and life table of the major vector of ZIKV, *Aedes* mosquitoes, are also well studied^{36, 37, 38, 39, 40, 41}, and can be incorporated in the compartments for vectors.

In this work, we present a compartmental model for ZIKV based on SEIR host model and SEI vector model, using logistic growth in human population and dynamic vector population⁴². We conduct sensitivity analysis on multiple parameters to understand the risk and transmission of ZIKV, and evaluate the efficiency of different containment strategies. We discuss how this may assist in decision making to control and mitigate outbreaks.

Method and Design

The compartmental model consists of two parts: the compartments for human, subscripted with H ; and compartments for vectors, subscripted with V . Although there have been a few reported deaths related to ZIKV, most ZIKV cases only result in mild symptoms among infectious individuals. We use an SEIR (Susceptible, Exposed, Infectious, Recovered) model to describe the disease spread among the human population. Let $S_H(t)$, $E_H(t)$, $I_H(t)$ and $R_H(t)$ denote the number of individuals in each compartment at time t . The human population at time t is given as $N_H(t) = S_H(t) + I_H(t) + E_H(t) + R_H(t)$. During the outbreak, the human population is assumed to follow a logistic growth with carrying capacity K_H .

The compartments for vectors consist of both the infectious status and the life stages of the mosquito. In particular, a typical life cycle of a mosquito has four stages: egg, larva, pupa, and adult. Let $V_1(t)$, $V_2(t)$ and $V_3(t)$ denote the number of eggs, larvae, and pupae of vectors in the environment. Since only female adults bite humans and transmit the virus, there are no infectious compartments for these three stages. Let $S_V(t)$, $E_V(t)$ and $I_V(t)$ denote the number of susceptible, exposed, and infectious female adults in the environment at time t , and denote the total female vector population as $N_V(t) = S_V(t) + I_V(t) + E_V(t)$. Due to their short life cycle, the birth and death rate of vector population is taken into consideration. The transmission from susceptible to infectious in both human and vector population is associated with mosquito bites. Figure 1 shows the transition diagram.

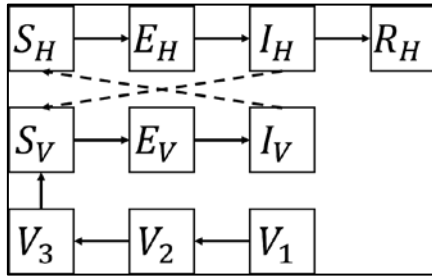


Figure 1. Transition diagram of the compartmental model. The dashed lines refer to the transitions associated with bites. The lines for vector birth and death are omitted for simplicity.

Let b denote the average biting rate per unit time of a female mosquito;; b typically ranges from 0.3 to 0.5 per day^{41, 43}. Let β represent the percentage of bites that transmit ZIKV. In addition, let M denote the amount of alternative hosts besides human in the environment. Then the number of bites received by a human per unit time is $bN_V/(N_H + M)$, among which I_V/N_V are from infectious mosquitoes. Therefore, the exposure rate of a human is given by

$$\beta \frac{bN_V}{N_H + M} \frac{I_V}{N_V} = \frac{\beta b I_V}{N_H + M}$$

Similarly, the exposure rate of a mosquito is $\beta b I_H/(N_H + M)$. Assume the inverse incubation time of ZIKV for human is α_H , and the recovery rate of an infectious individual is γ . The human population growth rate is λ , and the decreasing human population due to limited carrying capacity is $\lambda N_H^2/K_H$, and this amount is distributed in the four compartments according to their relevant size. Then the transitions between compartments for human can be described by the following system of ordinary differential equations:

$$\begin{aligned} \frac{d}{dt} S_H &= \lambda N_H \left(1 - \frac{S_H}{K_H}\right) - \frac{\beta b I_V}{N_H + M} S_H \\ \frac{d}{dt} E_H &= \frac{\beta b I_V}{N_H + M} S_H - \alpha_H E_H - \lambda N_H \frac{E_H}{K_H} \\ \frac{d}{dt} I_H &= \alpha_H E_H - \gamma I_H - \lambda N_H \frac{I_H}{K_H} \\ \frac{d}{dt} R_H &= \gamma I_H - \lambda N_H \frac{R_H}{K_H} \end{aligned}$$

For the vector compartments, let e_V denote the number of eggs laid per female mosquito per unit time, and α_V the inverse incubation time of ZIKV in vectors. Let τ_i and μ_i , $i \in \{1, 2, 3\}$ be the reciprocal of the development time and mortality rate of vectors in stage i , respectively. In addition, let μ_V denote the mortality rate of adults. Assume q is the percentage of females among adult vectors. Since mosquito larvae live in water with limited resources, the dynamics of the larvae population is also assumed to follow a logistic growth with carrying capacity K_V . The mortality rate of larvae caused by limited carrying capacity is denoted as κ . Then the transitions between compartments of vectors are described by the following system of ordinary differential equations:

$$\begin{aligned} \frac{d}{dt} V_1 &= e_V N_V - (\tau_1 + \mu_1) V_1 \\ \frac{d}{dt} V_2 &= \tau_1 V_1 - (\tau_2 + \mu_2) V_2 - \kappa \frac{V_2^2}{K_V} \\ \frac{d}{dt} V_3 &= \tau_2 V_2 - (\tau_3 + \mu_3) V_3 \\ \frac{d}{dt} S_V &= q \tau_3 V_3 - \frac{\beta b I_H}{N_H + M} S_V - \mu_V S_V \\ \frac{d}{dt} E_V &= \frac{\beta b I_H}{N_H + M} S_V - (\mu_V + \alpha_V) E_V \\ \frac{d}{dt} I_V &= \alpha_V E_V - \mu_V I_V \end{aligned}$$

Table 1. Parameters used in modeling the Zika disease dynamics

Name	Notation	Value	Literature Source
Number of infectious bites delivered per vector per day	βb	0.05 - 0.12	39, 40, 41, 43
Incubation time in human		2 - 4 days	44
Incubation time in vector		8 - 12 days	7
Recovery time	$1/\gamma$	5 - 7 days	45
Number of eggs laid per female mosquito per unit time		80 per day	37, 41
Egg development time		3 days	38, 41
Egg mortality		0.05	38, 41
Larval development time		9 days	38, 41
Larval mortality		0.025	38, 41
Pupal development time		3 days	41
Pupal mortality		0.0025	41
Adult mortality		0.05	38, 41
Female adult percentage	q	0.68	38

The system of ordinary differential equations above has a disease-free equilibrium if

$$e_V q \tau_1 \tau_2 \tau_3 > (\tau_1 + \mu_1)(\tau_2 + \mu_2)(\tau_3 + \mu_3) \mu_V$$

The number of noninfectious adult vectors at equilibrium if the inequality above holds is

$$S_V^* = \lim_{t \rightarrow \infty} S_V = \frac{e_V q \tau_1 \tau_2 \tau_3 - (\tau_1 + \mu_1)(\tau_2 + \mu_2)(\tau_3 + \mu_3) \mu_V}{(\tau_1 + \mu_1)(\tau_3 + \mu_3)^2 \mu_V^2 \kappa} K_V q \tau_2 \tau_3$$

The number of noninfectious humans at equilibrium follows the result of logistic growth, i.e.,

$$N_H^* = S_H^* = \lim_{t \rightarrow \infty} S_H = K_H$$

If we arrange the compartments corresponding to infectious stages as $\{E_H, I_H, E_V, I_V\}$, then the components of the next generation matrix^{46, 47} at equilibrium are given by:

$$F = \begin{pmatrix} 0 & 0 & 0 & \frac{\beta b S_H^*}{N_H^* + M} \\ 0 & 0 & 0 & 0 \\ 0 & \frac{\beta b S_V^*}{N_H^* + M} & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \quad V = \begin{pmatrix} \alpha_H + \frac{\lambda N_H^*}{K_H} & 0 & 0 & 0 \\ -\alpha_H & \gamma + \frac{\lambda N_H^*}{K_H} & 0 & 0 \\ 0 & 0 & \mu_V + \alpha_V & 0 \\ 0 & 0 & -\alpha_V & \mu_V \end{pmatrix}$$

Therefore, the basic reproduction number for our model is the spectral radius of the next generation matrix FV^{-1} :

$$R_0 = \rho(FV^{-1}) = \frac{\beta b}{(K_H + M)} \sqrt{\frac{\alpha_H \alpha_V K_H S_V^*}{(\alpha_H + \lambda)(\gamma + \lambda)(\alpha_V + \mu_V) \mu_V}}$$

To quickly contain the outbreak and reach a disease-free equilibrium, it is desirable to reduce the basic reproduction number such that $R_0 < 1$. In that case, an infected individual produces less than one new infected individual during its infectious period, thus the outbreak will start to contain. Using the R_0 formulation, we may alter the parameters by introducing interventions to contain the outbreak. In particular, the following two strategies are considered:

1. Reduce biting rate b by avoiding mosquito bites, using insect repellents, wearing long-sleeved clothes and long pants, and using air conditioning and window/door screens to keep mosquitoes outside.
2. Reduce adult female vector population or increase adult female vector mortality rate μ_V at equilibrium S_V^* by widely applying insecticides in areas with high (infectious) mosquito population, or introducing genetically modified *Aedes* mosquitoes, which will cause its offspring to die by reducing larval survival rate and adult longevity^{48, 49}.

To validate the performance of the model and investigate the effectiveness of containment strategies, we perform sensitivity and scenario analysis using the 2015 ZIKV data. For sensitivity analysis, we vary the recovery time (γ) and the incubation time of human (α_H). For containment strategies, we experiment with biting rate (b), adult vector mortality rate (μ_V), carrying capacity of larvae (K_V), and adult vector population (N_V). We report the estimated number of infections and time to reach containment. The simulation is performed until an asymptotic behavior is achieved. This signals containment is reached – the cumulative number of infections plateaus.

Results

Table 2 summarizes the parameters used in modeling the Zika disease dynamics for Rio de Janeiro State (Sentinel surveillance sites reports⁵⁰) and for a town with 100,000 people. Between January 2015 to July 2015, a total of 364

suspected cases were reported in Rio de Janeiro State. By November, the outbreak was contained with approximately 440 infection cases.

Table 2. Estimated parameter values used in the simulation.

Name	Notation	Rio de Janeiro State	A town of 100,000
Initial human population	$N_H(0)$	16,231,365	100,000
Initial human infections	$I_H(0)$	1	10
Initial adult vector population	$N_V(0)$	70,000,000	500,000
Initial exposed adult vectors	$E_V(0)$	5	1,000
Initial infectious adult vectors	$I_V(0)$	5	1,000
Initial number of eggs	$V_1(0)$	180,000,000	500,000
Initial number of larvae	$V_2(0)$	90,000,000	150,000
Initial number of pupae	$V_3(0)$	70,000,000	100,000
Number of infectious bites delivered per vector per day	βb	0.071	0.09
Carrying capacity for human	K_H	30,000,000	120,000
Carrying capacity for larvae	K_V	100,000,000	5,000,000
Larvae mortality rate due to limited carrying capacity	κ	1	1
Number of alternative hosts	M	100,000	80,000
Human population growth rate	λ	0.00005	0.00005
Recovery time of human	$1/\gamma$	7 days	7 days
Incubation time of human	$1/\alpha_H$	3 days	3 days
Incubation time of vector	$1/\alpha_V$	10 days	10 days
Basic reproduction number (estimate from parameters)	R_0	1.1139	1.109
Simulation start date		Jan 1, 2015	Day 0 (outbreak began)
Implementation of strategies		May 1, 2015	Day 100
Time for containment without intervention (when asymptotic behavior is observed)		Nov.2015	Day 900

Rio de Janeiro State, Brazil Figure 2 contrasts the reported cases versus simulation by varying the recovery time (γ) and the incubation time of human (α_H). The results suggest that Table 2 provides good estimates for the reported cases (green solid curve vs red dotted curve). When the recovery time increases from 6 to 8 days (dotted vs dashed green curves), the cumulative infections increases from 340 to 566 by July 2015 (66.5% increase). Longer recovery time also leads to delayed containment. This confirms that recovery time has significant impact on disease containment. Thus timely treatment and vaccination for combating ZIKV are critical.

Varying the human incubation period from 2 days to 4 days has a relatively minor influence on the overall infections (from 427 to 459, 7.5% increase) and the outbreak trend (contrast the three solid lines).

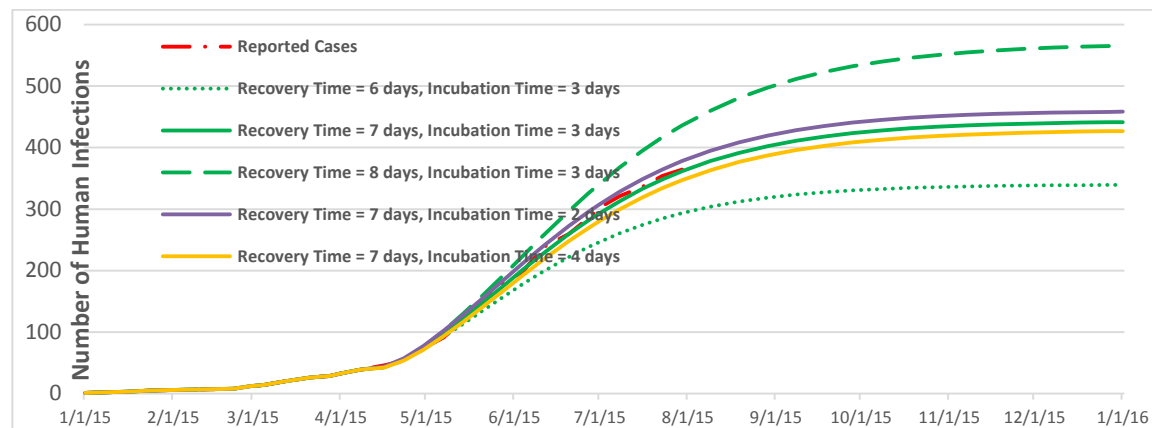


Figure 2. Varying recovery time and incubation time for human and its effect on overall infection and containment.

Figure 3 contrasts containment effectiveness for four strategies: reducing biting rate (b), increasing adult vector mortality rate (μ_V), reducing larvae carrying capacity (K_V), and reducing adult female vector population (N_V). Without these strategies, total infection reaches 443 and will be contained by October 2015. In our analysis, we assume each strategy is implemented on May 1, 2015. Figure 3(a) shows that reducing biting rate by a mere 20% will lead to an early containment by August 2015 (90 days after implementation) with 270 total infections (39% reduction). The containment is achieved rapidly (< 60 days after implementation) with total infection under 200 when biting rate is

reduced by at least 40%. This proves that reducing biting rate is highly effective. This strategy is also relatively easy to implement.

Increasing the mortality rate of adult vectors is also effective. Figure 3b shows that increasing the vector mortality rate by 20% will lead to an August containment with total infection of 282 (36% reduction). Containment will be achieved instantly with no more than 220 infections if the mortality rate is increased by at least 40%. While this strategy can be achieved by introducing genetically modified mosquitoes to the environment, it is difficult to realize and economically inferior when compared to other strategies.

The larvae carrying capacity has to be reduced by 80% in order for total infection to reduce to 330 (26% reduction) with containment achieved by October (Figure 3c). Nonetheless, examining and clearing the water ponds and humid areas where the larvae live remains an important strategy for public health protection.

Reducing vector population not only affects the number of infections at containment, it also changes the trend of the outbreak (Figure 3d). Specifically, reducing total adult vector population by 20% would postpone the outbreak by a month and reduce total infection to 334. Containment can be rapid when reduction is targeted at 80% (total infection reduces to 130). Figures 3c and 3d highlight that reducing adult population is more effective, though applying insecticide widely may come with numerous environmental and health issues.

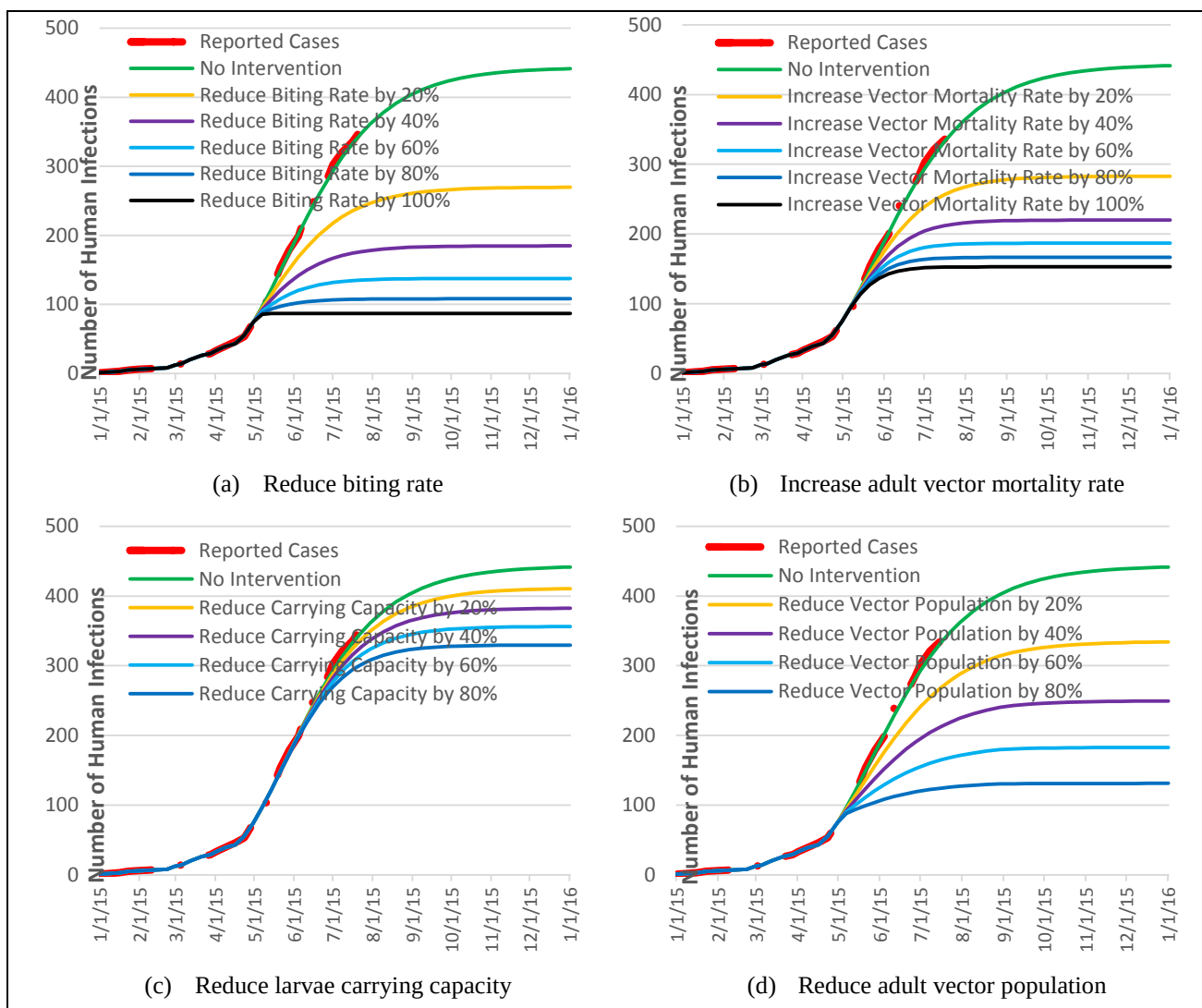


Figure 3. Contrast of total infection and outbreak trends for four containment strategies.

In practice, combination strategies are often most effective. For brevity, Figures 4a and 4b report containment results for combination of two strategies only. In both figures, the vertical axis represents the cumulative number of human infections at containment. When 40% reduction is achieved in both biting rate and larvae carrying

capacity, it results in 180 total infections (Figure 4a). Similarly, the combination of reducing vector population by 20% and increasing vector mortality rate by 20% results in 232 total infections (Figure 4b).

Since it can be difficult to achieve high effectiveness using a single strategy, combination strategies are appealing. Our results (Figures 4a and 4b) also display marginal improvement beyond certain threshold values in the strategies. This helps policy makers to invest their resources optimally for best possible gain.

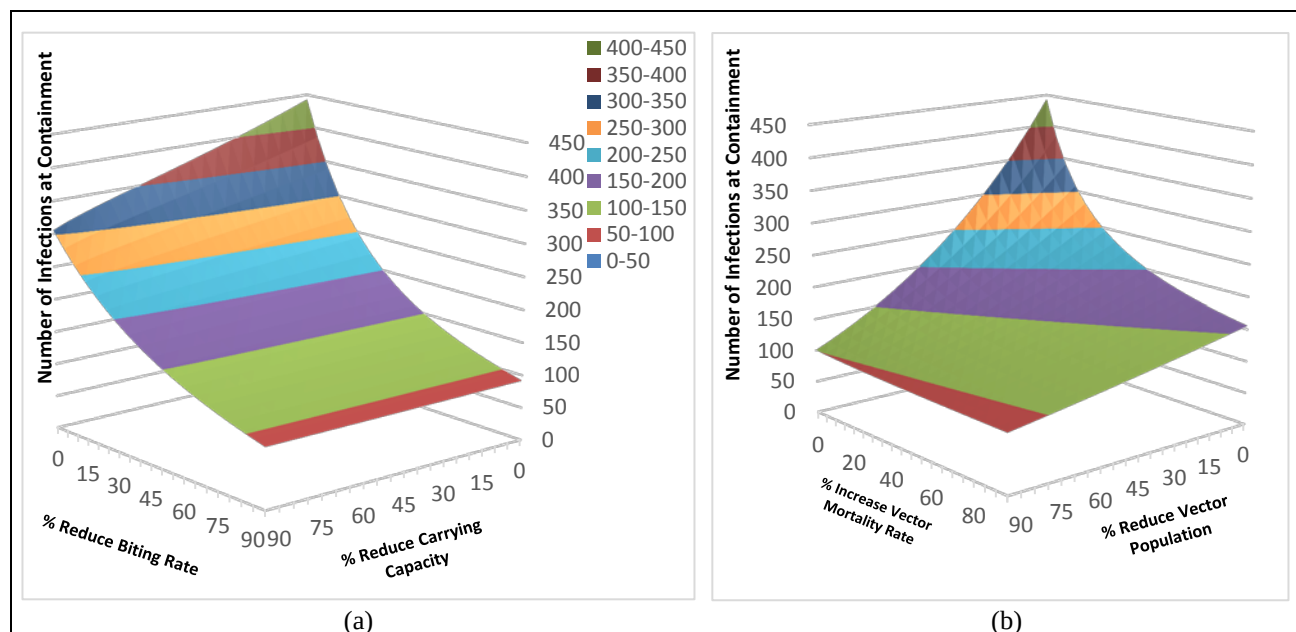


Figure 4. Cumulative number of human infections at containment under combination strategies.

A Town with 100,000 People To better understand the behavior and sensitivity of our model and the disease spread characteristics, we perform the analysis on a town with 100,000 people, using the parameters given in Table 2. Without any intervention, containment will occur at Day 900 with 42,000 overall infections.

We observe some similarities as in the Rio de Janeiro State analysis. An increase in human recovery time delays the containment and increases the total number of infections (Figure 5a). Similarly, the incubation period for humans has little effect on the overall disease trend (Figure 5c). On the other hand, the incubation time for the vectors significantly affects the overall number of infections while exhibiting the same outbreak pattern (Figure 5d).

We explore alternative hosts and their impact on the pandemic. Increasing the number of alternative hosts from 20,000 to 100,000 (Figure 5b) will slow the outbreak and reduce the total number of infections by 83.3%. This suggests that introducing alternative hosts has potential in mitigating an outbreak. However, it is possible that these hosts can get infected with ZIKV and spread it to humans via other means. The ecological and disease impact cannot be fully assessed at this point, thus rendering it an unsafe venue for ZIKV disease containment.

We assume each strategy is implemented on Day 100 after the outbreak. Figure 6 reports containment effectiveness for each strategy. Reducing bite rate by 20% (Figure 6a) and increasing vector mortality by 20% (Figure 6b) have similar effectiveness in containing the disease spread (Day 500 vs Day 300) and with comparable overall infection (12,000 vs 11,000). Both strategies are very effective – reducing infections by over 70%.

Reducing larvae carrying capacity has a significant impact on the outbreak (Figure 6c). In particular, over 40% reduction in infection is observed when 20% reduction in carrying capacity is in effect. On the other hand, reducing vector population is not as effective in containing the ZIKV outbreak (Figure 6d). Nonetheless, reducing vector population does influence the pandemic course: when the vector population is reduced by 20%, it postpones the onset of the outbreak by approximately 100 days, and reduces the total infection by 9.5%. These results are different from the findings in the Rio de Janeiro State analysis where reducing the carrying capacity of larvae proves to be not very effective.

Figure 7 shows the combination strategies that can be implemented in practice in a cost-effective manner. Figure 7b shows the marginal gain in infection control through reduction in vector population.

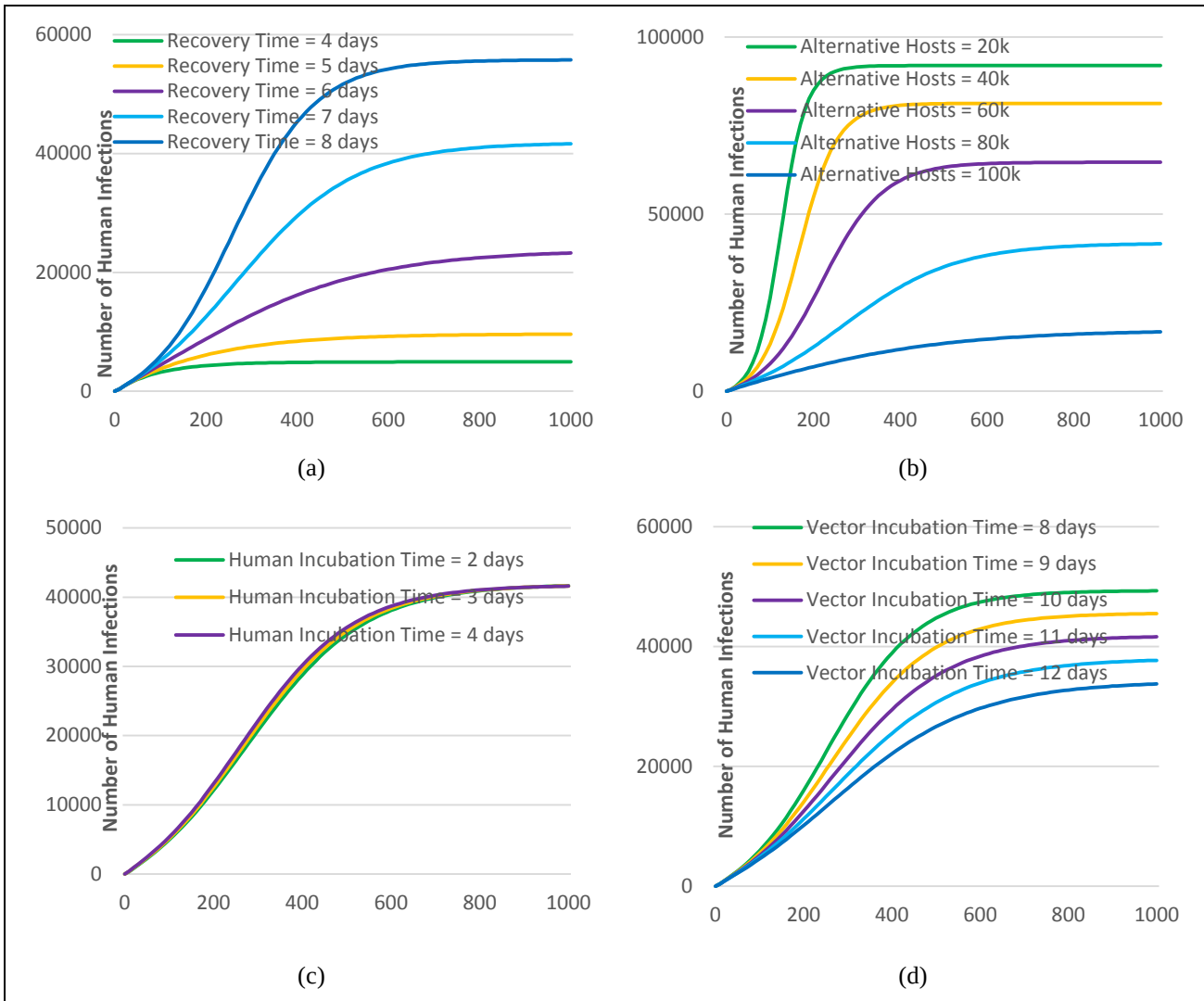
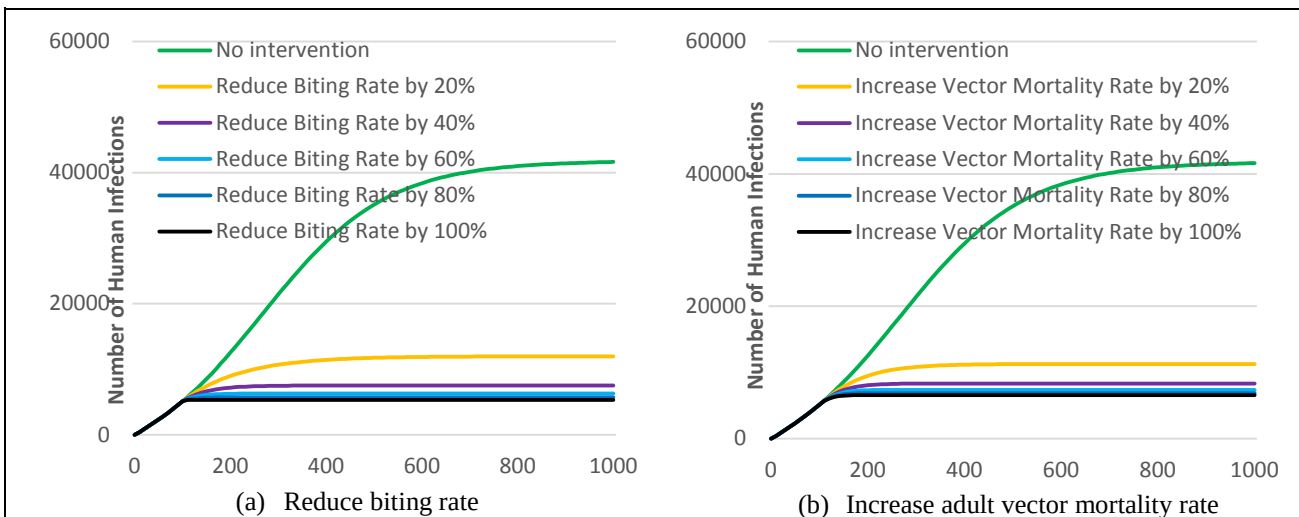


Figure 5. Varying recovery time, alternative hosts, and incubation time for human and vector and their effect on overall infection and containment.



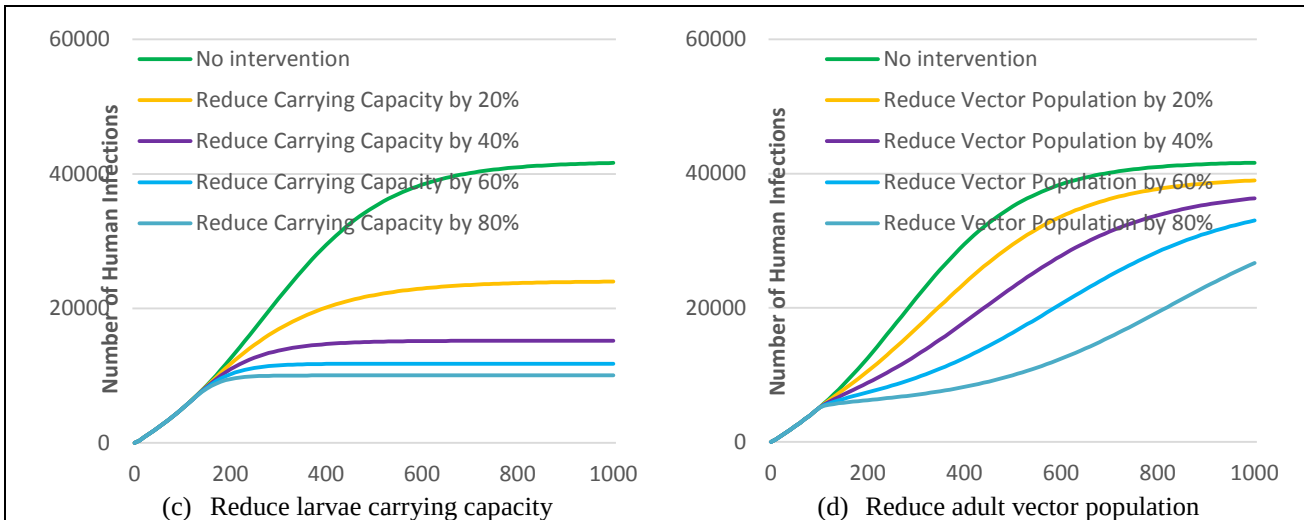


Figure 6. Contrast of total infection and outbreak trends for four containment strategies.

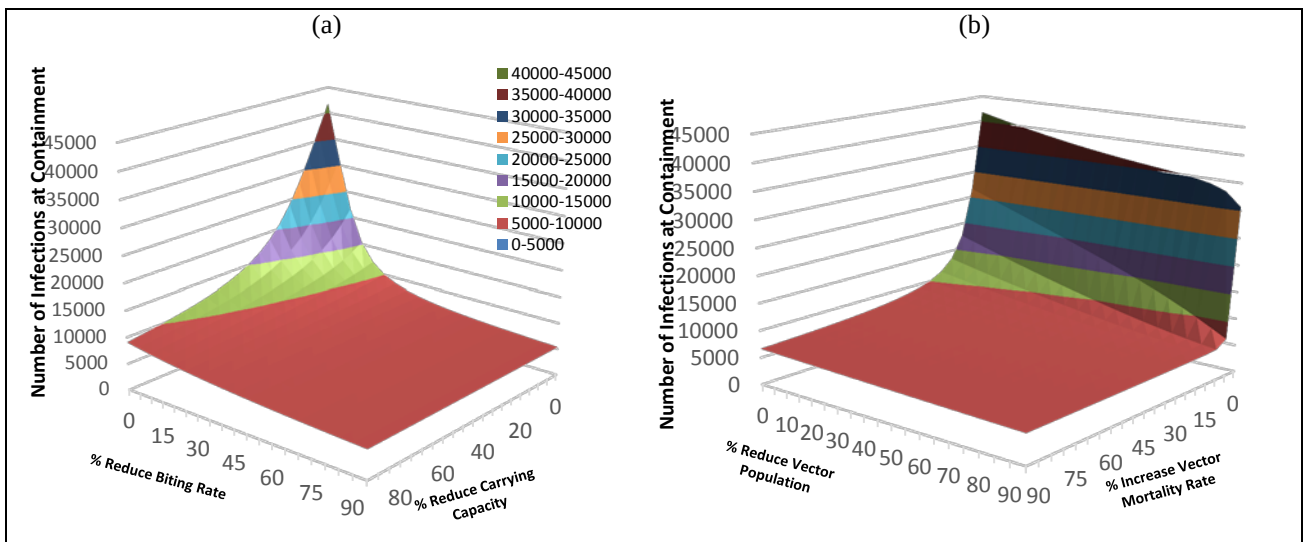


Figure 7. Cumulative number of human infections at containment under combination strategies.

Discussion

This work demonstrates the key importance of “digital disease surveillance” in response to waves of epidemics including ZIKV, Dengue, Ebola and cholera. Specifically, we develop a compartmental model with a vector-host structure utilizing logistic growth for human population and a dynamic vector population describing its full life cycle. We analyze the development of ZIKV outbreak and explore key parameters that influence epidemic severity. We study the effect of ‘average’ recovery time (of human) from infection, the number of alternative hosts, and the ZIKV incubation time in both vectors and humans. The analyses are performed on two populations with real data from Rio de Janeiro State. Our results reveal that the outbreak is sensitive to the first three parameters while human incubation period has only marginal impact.

We examine four containment strategies: reducing biting rate, increasing vector mortality rate, reducing larvae carrying capacity, and reducing vector population. All four strategies are effective (in varying degrees) in containing the outbreak and reducing the overall infection. While findings for reducing biting rate and increasing vector mortality rate are similar for both populations (both are superior to other strategies), results for the larvae carrying capacity and reduction in vector population differ. These findings suggest environmental and demographic information should be considered when determining proper containment strategies. Our study also shows that reducing vector population early on can help delay the onset of the outbreak.

Combination strategies are both promising in practice and cost-effective in achieving early containment. The model provides a decision support framework for policy makers to estimate the cost-effectiveness for each prevention

measure. Public health departments should select a strategy portfolio compatible with their local environment and regional demographics. In addition, the public should be educated and informed of ZIKV status. Population behaviors (protecting themselves by reducing biting, cleaning water ponds to rid of larvae, etc.) have demonstrably significant impact on containing and mitigating the outbreak.

The multiple components involving the dynamics of human and vector populations in this model allow flexibility in characterizing disease spread and performing strategic analysis. However, obtaining/determining all essential input parameters for this to be practical may prove difficult. The model can be simplified without diminishing its quality and rigor in disease dynamics and infection/containment outcome prediction⁵¹. Such a model may be desirable when detailed input data are not readily available.

For example, the natural human population can be assumed to be constant throughout the outbreak since the magnitude of natural population growth is relatively small within the pandemic period. The mosquito population lifecycle can also be condensed to a simple birth-death process. In Lee et al 2016⁵¹, we propose a simplified model with constant human and mosquito population to study the ZIKV outbreak. The study predicts the disease trends and characterizes containment strategies for Brazil and Puerto Rico. The model has accurately predicted ZIKV outbreak trends for these two countries. It predicts the number of infectious pregnant women in Puerto Rico within 5% of the actual documented numbers (as of July 2016)⁵¹. Although the simplified model is statistically useful and usable, additional components and parameters presented herein should help decision makers better understand the details of ZIKV outbreaks. It may also provide them with more analytic-based options in deciding the proper containment strategies when such input data are available.

Modern information technologies which are capable of tracking disease spread and vector activities play an indispensable role in supporting disease modeling and in mitigating and containing epidemiology outbreaks. Informatics applications such as social media and internet-based participatory surveillance systems have already demonstrated their capability in helping us understand and prevent disease outbreaks like Dengue and influenza^{52,53}. Software and hardware such as ArcGIS platform and mobile DNA sequencing lab are also designed to help track the current ZIKV outbreak. These technologies capture valuable inputs for our model, which is very flexible and can be easily adapted in real-time as real data feeds arrive. With the support of these technologies, the model parameters can be estimated accurately and updated on a regular basis. The model output can then be used to predict the status of the outbreak in a timely manner. It also reflects any changes in the environment, policy, and human and social behavior. Conversely, the model can also serve as a guideline of the application of these decision technologies such that they are deployed and utilized optimally by policy makers. Informatics can help us understand the underlying mechanism of the pandemic spread. It also enables decision and policy makers to understand/update the planning and response strategy in real time to maximize the containment effect while minimizing the overall infection / mortality and the associated costs and health burden.

There are limitations to our study. We assume a homogeneous human population in the model, which may not hold in practice. The possibility of contacting mosquitoes and the chance of getting infected from an infectious vector may differ among individuals and the environment. This can be overcome by creating additional compartments for each homogeneous subgroup. We have incorporated this subgroup concept in our vaccine prioritization analysis with great success²⁵. However, this requires even more parameter estimation and may lead to over-fitting. In addition, our conclusions are strongly influenced by results from a particular demographic and geographic profile. Local decision makers should input proper (best estimate) parameters into the model in order to conduct accurate analyses.

The current model does not include any treatment or vaccination effects since they are currently under developed. With deeper understanding of ZIKV and the presence of medications and vaccines, these factors can also be incorporated under our modeling framework.

Acknowledgement The work is partially supported by a grant from the National Science Foundation and the Centers for Disease Control and Prevention.

Reference:

1. Dick GW, Kitchen SF, Haddow AJ. Zika virus (I). Isolations and serological specificity. *Trans R Soc Trop Med Hyg.* 1952;46(5):509-20.
2. MacNamara FN. Zika virus: a report on three cases of human infection during an epidemic of jaundice in Nigeria. *Trans R Soc Trop Med Hyg.* 1954;48(2):139-45.
3. Dick GW. Zika virus (II). Pathogenicity and physical properties. *Trans R Soc Trop Med Hyg.* 1952;46(5):521-34.
4. Simpson DI. Zika virus infection in man. *Trans R Soc Trop Med Hyg.* 1964;58(4):339-48.
5. Adekolu-John EO, Fagbami AH. Arthropod-borne virus antibodies in sera of residents of Kainji Lake Basin, Nigeria 1980. *Trans R Soc Trop Med Hyg.* 1983;77(2):149-51.
6. Duffy MR, Chen TH, Hancock WT, Powers AM, Kool JL, Lanciotti RS et al. Zika virus outbreak on Yap Island, federated states of Micronesia. *N Engl J Med.* 2009;360(24):2536-43.
7. Hayes EB. Zika virus outside Africa. *Emerg Infect Diseases.* 2009;15(9):1347-50.
8. Pond WL. Arthropod-borne virus antibodies in sera from residents of South-East Asia. *Trans R Soc Trop Med Hyg.* 1963;57(5):364-71.
9. Cao-Lormeau V. Zika virus, French Polynesia, South Pacific, 2013. *Emerg Infect Diseases.* 2014;20(11):1960.

10. Campos GS, Bandeira AC, Sardi SI. Zika virus outbreak, Bahia, Brazil. *Emerg Infect Diseases*. 2015;21(10):1885.
11. Corsica F. Zika virus transmission from French Polynesia to Brazil. *Emerg Infect Diseases*. 2015;21(10):1887.
12. Camacho E, Paternina-Gomez M, Blanco PJ, Osorio JE, Aliota MT. Detection of autochthonous Zika virus transmission in Sincelejo, Colombia. *Emerg Infect Diseases*. 2016;22(5).
13. Grard G, Caron M, Mombo IM, Nkoghe D, Ondo SM, Jiolle D et al. Zika virus in Gabon (Central Africa)–2007: a new threat from *Aedes albopictus*? *PLoS Negl Trop Dis*. 2014;8(2):e2681.
14. Foy BD, Kobylinski KC, Chilson Foy JL, Blitvich BJ, Travassos da Rosa A, Haddow AD et al. Probable non-vector-borne transmission of Zika virus, Colorado, USA. *Emerg Infect Diseases*. 2011;17(5):880-2.
15. Besnard M, Lastère S, Teissier A, Cao-Lormeau V, Musso D. Evidence of perinatal transmission of Zika virus, French Polynesia, December 2013 and February 2014. *Euro Surveill*. 2014;19(13):20751.
16. Iosifidis S, Mallet HP, Goffart IL, Gauthier V, Cardoso T, Herida M. Current Zika virus epidemiology and recent epidemics. *Med Mal Infect*. 2014;44(7):302-7.
17. Wolfe ND, Kilbourn AM, Karesh WB, Rahman HA, Bosi EJ, Cropp BC et al. Sylvatic transmission of arboviruses among Bornean orangutans. *Am J Trop Med Hyg*. 2001;64(5):310-6.
18. Haddow AD, Schuh AJ, Yasuda CY, Kasper MR, Heang V, Huy R et al. Genetic characterization of Zika virus strains: geographic expansion of the Asian lineage. *PLoS Negl Trop Dis*. 2012;6(2):e1477.
19. Oliveira Melo AS, Malinger G, Ximenes R, Szejnfeld PO, Alves Sampaio S, Bispo de Filippis AM. Zika virus intrauterine infection causes fetal brain abnormality and microcephaly: tip of the iceberg? *Ultrasound Obstet Gynecol*. 2016;47(1):6-7.
20. Ventura CV, Maia M, Bravo-Filho V, Góis AL, Belfort R. Zika virus in Brazil and macular atrophy in a child with microcephaly. *Lancet*. 2016;387(10015):228.
21. Oehler E, Watrin L, Larre P, Leparc-Goffart I, Lastère S, Valour F et al. Zika virus infection complicated by Guillain-Barre syndrome—case report, French Polynesia, December. *Euro Surveill*. 2014;19(9):20720.
22. Bogoch II, Brady OJ, Kraemer MU, German M, Creatore MI, Kulkarni MA et al. Anticipating the international spread of Zika virus from Brazil. *Lancet*. 2016;387(10016):335-336.
23. Kutsuna S, Kato Y, Takasaki T, Moi M, Kotaki A, Uemura H et al. Two cases of Zika fever imported from French Polynesia to Japan, December 2013 to January 2014. *Euro Surveill*. 2014;19(4):1-5.
24. Kwong JC, Druce JD, Leder K. Zika virus infection acquired during brief travel to Indonesia. *Am J Trop Med Hyg*. 2013;89(3):516-7.
25. Lee EK, Yuan F, Pietz FH, Benecke BA, Burel G. Vaccine prioritization for effective pandemic response. *Interfaces*. 2015;45(5):425-43.
26. Lee EK, Chen CH, Pietz F, Benecke B. Disease propagation analysis and mitigation strategies for effective mass dispensing. *AMIA Annu Symp Proc*. 2010;427-31.
27. Grassly NC, Fraser C. Mathematical models of infectious disease transmission. *Nat Rev Microbiol*. 2008;6(6):477-87.
28. Longini IM, Ackerman E, Elveback LR. An optimization model for influenza A epidemics. *Math Biosci*. 1978;38(1-2):141-57.
29. Kuno G, Chang GJ, Tsuchiya KR, Karabatsos N, Cropp CB. Phylogeny of the genus *Flavivirus*. *J Virol*. 1998;72(1):73-83.
30. Feng Z, Velasco-Hernández JX. Competitive exclusion in a vector-host model for the dengue fever. *J Math Biol*. 1997;35(5):523-44.
31. Esteva L, Vargas C. A model for dengue disease with variable human population. *J Math Biol*. 1999;38(3):220-40.
32. Esteva L, Vargas C. Analysis of a dengue disease transmission model. *Math Biosci*. 1998;150(2):131-51.
33. Esteva L, Matias M. A model for vector transmitted diseases with saturation incidence. *J Biol Syst*. 2001;9(04):235-45.
34. Derouich M, Boutayeb A. Dengue fever: Mathematical modelling and computer simulation. *Appl Math Comput*. 2006;177(2):528-44.
35. Erickson RA, Presley SM, Allen LJ, Long KR, Cox SB. A dengue model with a dynamic *Aedes albopictus* vector population. *Ecol Model*. 2010;221(24):2899-908.
36. Dye C. Models for the population dynamics of the yellow fever mosquito, *Aedes aegypti*. *J Anim Ecol*. 1984;53(1):247-68.
37. Costero A, Edman JD, Clark GG, Scott TW. Life table study of *Aedes aegypti* (Diptera: Culicidae) in Puerto Rico fed only human blood versus blood plus sugar. *J Med Entomol*. 1998;35(5):809-13.
38. Sowilem MM, Kamal HA, Khater EI. Life table characteristics of *Aedes aegypti* (Diptera: Culicidae) from Saudi Arabia. *Trop Biomed*. 2013;30(2):301-14.
39. Vazeille M, Yébakima A, Lourenço-de-Oliveira R, Andriamahefazafy B, Correia A, Rodrigues J et al. Oral receptivity of *Aedes aegypti* from Cape Verde for yellow fever, dengue, and chikungunya viruses. *Vector Borne Zoonotic Dis*. 2013;13(1):37-40.
40. Vazeille M, Rosen L, Mousson L, Failloux AB. Low oral receptivity for dengue type 2 viruses of *Aedes albopictus* from Southeast Asia compared with that of *Aedes aegypti*. *Am J Trop Med Hyg*. 2003;68(2):203-8.
41. Erickson R. Impact of climate change on the population dynamics of *Aedes albopictus* and disease dynamics of dengue. [Master's Thesis]. Texas Tech University; 2009.
42. Erickson RA, Presley SM, Allen LJ, Long KR, Cox SB. A stage-structured, *Aedes albopictus* population model. *Ecol Model*. 2010;221(9):1273-82.
43. Gubler D. Dengue. In: Monath T., ed. *The Arboviruses: Epidemiology and Ecology*, vol. 2. Boca Raton, FL: CRC Press; 1986.
44. Bearcroft WG. Zika virus infection experimentally induced in a human volunteer. *Trans R Soc Trop Med Hyg*. 1956;50(5):438-41.
45. CDC: Zika Virus. Available at: <http://www.cdc.gov/zika/>. Accessed March 2, 2016.
46. Van den Driessche P, Watmough J. Reproduction numbers and sub-threshold endemic equilibria for compartmental models of disease transmission. *Math Biosci*. 2002;180(1):29-48.
47. Diekmann O, Heesterbeek JA, Metz JA. On the definition and the computation of the basic reproduction ratio R_0 in models for infectious diseases in heterogeneous populations. *J Math Biol*. 1990;28(4):365-82.
48. Carvalho DO, McKemey AR, Garziera L, Lacroix R, Donnelly CA, Alphey L, Malavasi A et al. Suppression of a field population of *Aedes aegypti* in Brazil by sustained release of transgenic male mosquitoes. *PLoS Negl Trop Dis*. 2015;9(7):e0003864.
49. Bargielowski I, Nimmo D, Alphey L, Koella JC. Comparison of life history characteristics of the genetically modified OX513A line and a wild type strain of *Aedes aegypti*. *PLoS One*. 2011;6(6):e20699.
50. Brasil P, Calvet G, Siqueira A et al. Zika virus outbreak in Rio de Janeiro, Brazil: clinical characterization, epidemiological and virological aspects. *PLoS Negl Trop Dis*. 2016;10(4):e0004636.
51. Lee EK, Liu Y, Pietz FH. Analyzing Strategies for Containing the Zika Outbreak. Selected to present at AAAS 2017. Submitted to PNAS.
52. Lwin M, Vijaykumar S, Fernando O et al. A 21st century approach to tackling dengue: Crowdsourced surveillance, predictive mapping and tailored communication. *Acta Trop*. 2014;130:100-107.
53. Wójcik O, Brownstein J, Chunara R, Johansson M. Public health for the people: participatory infectious disease surveillance in the digital age. *Emerg Themes Epidemiol*. 2014;11(1):7.